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LncRNA MIR210HG promotes the proliferation of colon cancer cells by inhibiting ferroptosis through binding to PCBP1

Xiao-Qian Wang[1,2], A.-Qiang Fan[1,2] & Liu Hong[1]✉

This study aimed to investigate the role of the MIR210 host gene (MIR210HG), a long noncoding RNA (lncRNA), in the proliferation of colon cancer cells and its potential mechanism involving the ferroptosis pathway. We assessed MIR210HG expression in colon cancer cell lines and tissues, and examined the effects of its overexpression and knockdown on cell proliferation. Proteomic analysis was conducted to explore the interaction between MIR210HG and ferroptosis pathway components. The binding of MIR210HG to poly(rC) binding protein 1 (PCBP1) was predicted using catRAPID and confirmed through RNA pull-down and RNA immunoprecipitation (RIP) experiments. MIR210HG was significantly upregulated in colon cancer cells and tissues. Its overexpression promoted, while its knockdown inhibited, colon cancer cell proliferation. MIR210HG was found to be associated with ferroptosis pathway components and to bind to PCBP1, which was experimentally validated. The inhibition of ferroptosis by MIR210HG through PCBP1 binding was confirmed, highlighting its role in promoting cell proliferation. MIR210HG promotes colon cancer cell proliferation by binding to PCBP1 and inhibiting ferroptosis. These findings suggest MIR210HG as a potential therapeutic target for colon cancer.

Keywords Long noncoding RNA, Ferroptosis, PCBP1, Colon cancer, Proliferation, MIR210HG

According to the International Agency for Research on Cancer (IARC) of the World Health Organization, it is projected that in 2020, the incidence of colon cancer (CC) in China will surpass 300,000 new cases, with a mortality rate exceeding 160,000[1]. This malignancy represents a substantial peril to the well-being of the Chinese population. The clinical stage of the disease has emerged as the most crucial prognostic factor for survival prediction among CC patients, with notable variations observed in therapeutic outcomes across different stages[2,3]. According to colorectal cancer statistics from 2012 to 2018 in the United States, the five-year relative survival rate for early-stage patients can reach up to 91%, whereas for patients with distant metastasis, the rate is only 14%[4]. The escalating growth pattern in advanced cases is a matter of concern. A thorough investigation of the mechanisms underlying the proliferation and metastasis of CC cells can contribute to a comprehensive understanding of their biological characteristics and facilitate the identification of novel biological targets for the early detection and prevention of metastasis.

Recent research has demonstrated the significant involvement of long noncoding RNAs (lncRNAs) in the pathogenesis of CC, effectively augmenting the gene regulatory machinery underlying this disease. lncRNAs exert their biological functions through diverse mechanisms, including transcriptional and translational regulation[5,6], protein modification[7–9], and the formation of RNA-protein[10,11] or RNA-DNA complexes[12,13]. Moreover, lncRNAs play pivotal roles in various disease processes, notably cancer[14–16].

The lncRNA MIR210HG, the host gene encoding miR210[17], is located at 11p15.5 and spans a length of 2303 nucleotides[18]. Extensive research has demonstrated that MIR210HG exhibits oncogenic properties, facilitating the advancement of tumors across multiple cancer types, such as glioma[19], glioblastoma[20,21], breast cancer[22–24], cervical cancer[25–27], endometrial cancer[28], gastric cancer[29], colon cancer[30], hepatocellular carcinoma[31,32], hepatoblastoma[33], and non-small cell lung cancer[34].

In nonneoplastic diseases, the hypoxia-induced long noncoding RNA MIR210HG facilitates the expression of HIF1-α by suppressing miR-93-5p in renal tubular cells[35]. MIR210HG is abnormally expressed in the seminal plasma of varicocele patients and has been linked to varicocele-associated oligoasthenozoospermia[36] and

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male infertility. Furthermore, MIR210HG exacerbates the sepsis-induced inflammatory response of proximal tubular epithelial cells through the NF-κB signaling pathway[37]. Notably, MIR210HG has been identified as a novel biomarker for diagnosing preeclampsia[39] and displays seasonal variation in its expression within full-term human placentas[40]. However, the precise molecular mechanisms and regulatory functions of MIR210HG in relation to CC remain unknown.

Results

Upregulation of MIR210HG is associated with disease-free survival in CC patients

To investigate the involvement of MIR210HG in CC, we examined its expression levels in both CC tissues and normal tissues from the TCGA database. The analysis revealed the upregulation of MIR210HG in CC tissues, as depicted in Fig. 1A. This finding was subsequently validated through quantitative polymerase chain reaction (q-PCR) analyses performed on clinical samples ($n=20$), as illustrated in Fig. 1C. Furthermore, an investigation was conducted to assess the expression levels of MIR210HG in various CC cell lines, namely, the HCT116, SW620, HT29, LOVO, and SW480 cell lines, as well as the human normal colon epithelial cell line FHC. The findings revealed a noteworthy upregulation of MIR210HG expression in the five CC cell lines compared to the FHC cell line (Fig. 1B). These findings suggest that MIR210HG is significantly overexpressed in CC. As illustrated in Fig. 1D of the TCGA dataset, the overexpression of MIR210HG was significantly associated with both overall survival and disease-free survival, indicating a potential close relationship between MIR210HG overexpression and the onset and progression of colon cancer in patients.

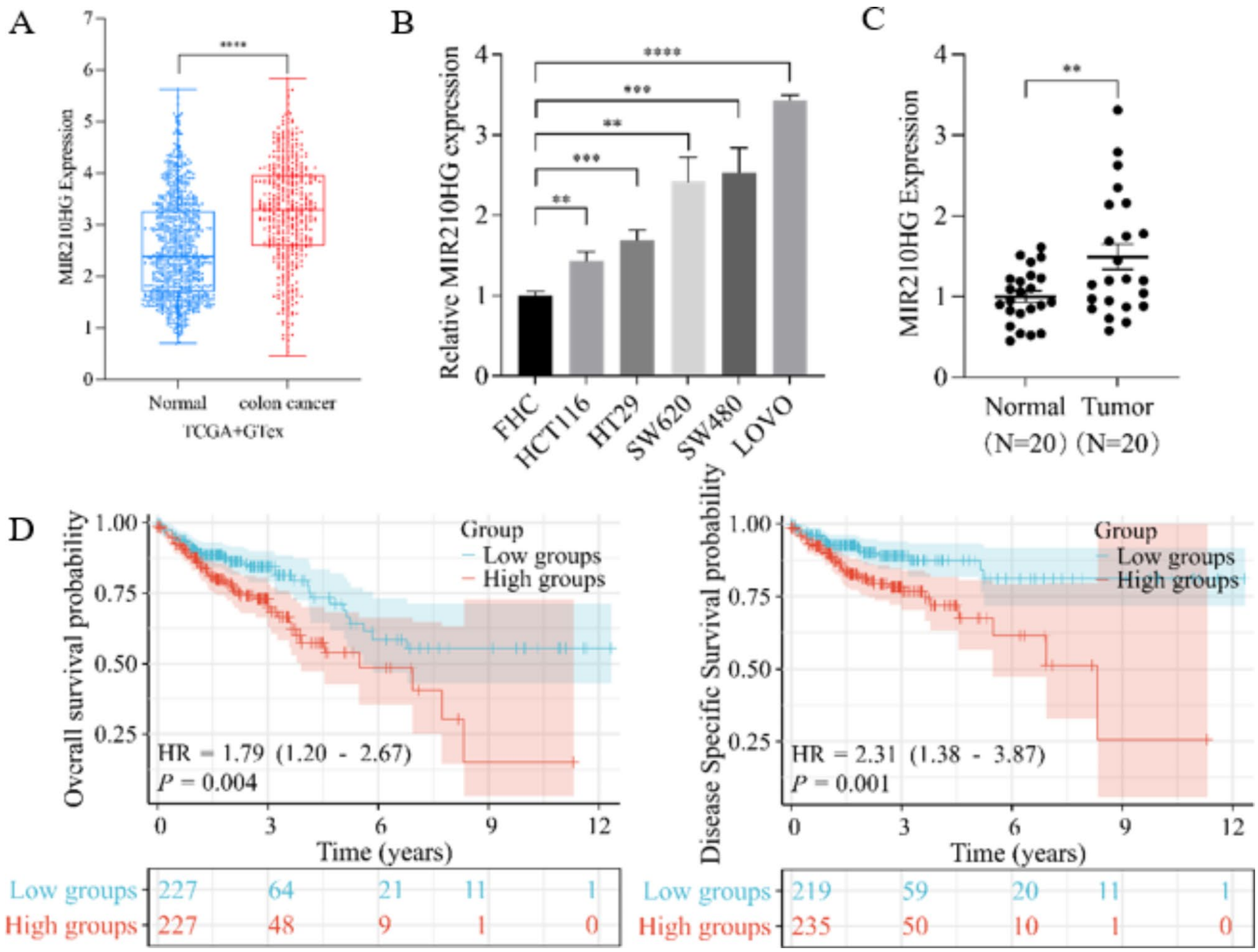

Fig. 1. The distribution of MIR210HG gene expression in tumor tissues and normal tissues was analyzed using publicly available database downloads (**A**) and clinically collected tissue samples (**C**). (**B**) qRT-PCR was utilized to detect the expression of MIR210HG in various colon cancer cell lines (HCT116, SW620, HT29, LOVO, and SW480) and the human normal colon epithelial cell line FHC. The abscissa represents different groups of samples, and the ordinate represents the expression distribution of the genes. The top represents significant p values, $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$, asterisks (*) represent significant differences. The significance of differences between two groups was compared through the Wilcoxon test. One-way ANOVA was used to compare multiple groups. (**D**) Kaplan-Meier survival maps were generated for cancer patients with high and low MIR210HG expression using TCGA data.

Upregulation of MIR210HG promotes CC cell proliferation in vitro and in vivo

The objective of this study was to investigate the impact of MIR210HG on the proliferation of CC cells. This was achieved by transfecting SW480 and SW620 cells with lentiviral vectors and antisense oligonucleotides (ASOs) to establish overexpression and knockdown cell lines, as shown in Fig. 2A. The results of the CCK-8 proliferation assay, as depicted in Fig. 2B, indicate that increased expression of the MIR210HG gene significantly promotes the proliferation of SW480 and SW620 cells, while decreased expression of the MIR210HG gene leads to a reduction in cell proliferation in these cell lines. The findings from the plate colony formation assay provided evidence that the overexpression of the MIR210HG gene significantly enhanced colony formation in SW480 and SW620 cells, as depicted in Fig. 2C. To facilitate a more in-depth investigation of the function of MIR210HG in vivo, an animal model was developed through subcutaneous injection of male nude mice with the SW480 cell line that overexpresses MIR210HG. The findings revealed a significant increase in tumor volume following the overexpression of MIR210HG (Fig. 2D-E). Significantly, we conducted HE staining and IHC detection of Ki-67 expression in murine tumor tissues. Remarkably, an increase in Ki-67 expression was detected in mice subjected to subcutaneous injection of SW480 cells overexpressing MIR210HG (Fig. 2E). Notably, the overexpression of MIR210HG substantially promoted the growth of CC in vivo. These findings strongly imply that MIR210HG potentially promotes the proliferation of CC cells.

MIR210HG promotes CC cell proliferation by inhibiting the ferroptosis pathway

To determine the biological role of MIR210HG in CC cells, protein was extracted from SW620 cells overexpressing MIR210HG, and subsequent protein profiling analysis was performed to compare the overexpression group to the control group. The analysis revealed a total of 1986 upregulated proteins and 4848 downregulated proteins among the differentially expressed proteins compared to the control proteins (Fig. 3A,B). The results of GO (Fig. 3C) and KEGG (Fig. 3D) analyses demonstrated that the overexpression of MIR210HG has the potential to inhibit the ferroptosis pathway. Furthermore, the significant increase in GPX4 expression in the total protein fraction of both SW480 and SW620 cells overexpressing MIR210HG further supported the inhibitory effect of MIR210HG overexpression on ferroptosis (Fig. 3E). Transmission electron microscopy (TEM) was employed to examine the morphological features of SW620 cells overexpressing MIR210HG (Fig. 3F). The results revealed that control cells exhibited diminished mitochondrial volume, increased bilayer membrane density, and a decreased or absent mitochondrial cristae, which aligns with the distinctive mitochondrial morphology associated with ferroptosis. Furthermore, the upregulation of MIR210HG mitigated the subcellular morphological alterations associated with ferroptosis in CC cells.

MIR210HG inhibits the ferroptosis pathway by interacting with PCBP1

The catRAPID omics algorithm (http://service.tartaglialab.com/page/catrapid_group) was used to investigate lncRNA-protein interactions. The catRAPID omics algorithm yielded a prediction of 2064 downstream binding proteins (Table S1), and 33 ferroptosis-related proteins were identified from the KEGG website (https://www.kegg.jp/) (Table S2). The intersection of these two sets of proteins revealed potential ferroptosis-related proteins that could bind downstream of MIR210HG (Fig. 4A). According to the z value analysis, compared with the other three proteins, PCBP1 exhibited the most pronounced binding affinity for MIR210HG. MIR210HG potentially inhibits the ferroptosis pathway by binding to PCBP1. To validate this hypothesis, RNA pull-down experiments (Fig. 4B) and RIP (Fig. 4C) experiments were conducted. In the RNA pull-down experiment, MIR210HG was transcribed into four segments (sense1–4) in vitro and subjected to separate pull-down experiments. The silver staining results revealed a significant differential band in the Sense-2 Lane, indicating a good overall pull-down enrichment effect. Both the Sense-2 and Input groups exhibited bands corresponding to the target protein PCBP1, suggesting that MIR210HG can bind to the PCBP1 protein. RIP-qPCR analysis demonstrated a notable increase in MIR210HG enrichment within cells in the IP experimental group compared with the IgG control group, suggesting a potential interaction between PCBP1 and MIR210HG.

PCBP1 serves as the primary cytoplasmic iron chaperone protein in mammalian cells[41–43], facilitating the metallization and storage of iron into ferritin in conjunction with reduced glutathione (GSH)[44,45]. Additionally, iron chaperone proteins play a crucial role in mitigating the instability of iron pools (LIPs)[46,47], safeguarding proteins and organelles against detrimental reactive oxygen species (ROS), and regulating iron efflux through ferritin[48,49].

To investigate the role of PCBP1 in facilitating colon cancer cell proliferation through MIR210HG, we established cell models of MIR210HG overexpression and MIR210HG overexpression combined with PCBP1 knockdown in SW480 cells (Fig. 5A). Our findings from the CCK-8 assay (Fig. 5B) and colony formation assay (Fig. 5C) indicated that PCBP1 knockdown significantly attenuated the proliferative effect of MIR210HG on colon cancer cells, with the results being statistically significant.

To ascertain the inhibitory role of the iron chaperone protein PCBP1 in ferroptosis in colon cancer cells, we established a series of cellular models, namely, a control group, a PCBP1-knockdown group, and a PCBP1-restored expression group, using the SW480 colon cancer cell line (Fig. 6A). Next, we exposed these three groups of cells to the ferroptosis inducer erastin, with a solvent control group also included, and quantified the intracellular iron content before and after treatment (Fig. 6B). The experimental results revealed that erastin significantly enhanced the total iron burden within colon cancer cells. Notably, the knockdown of PCBP1 further potentiated the accumulation of total iron within colon cancer cells, whereas the re-establishment of PCBP1 expression mitigated the total cellular iron levels. These findings suggest that PCBP1 modulates the sensitivity of colon cancer cells to ferroptosis by regulating intracellular iron metabolism.

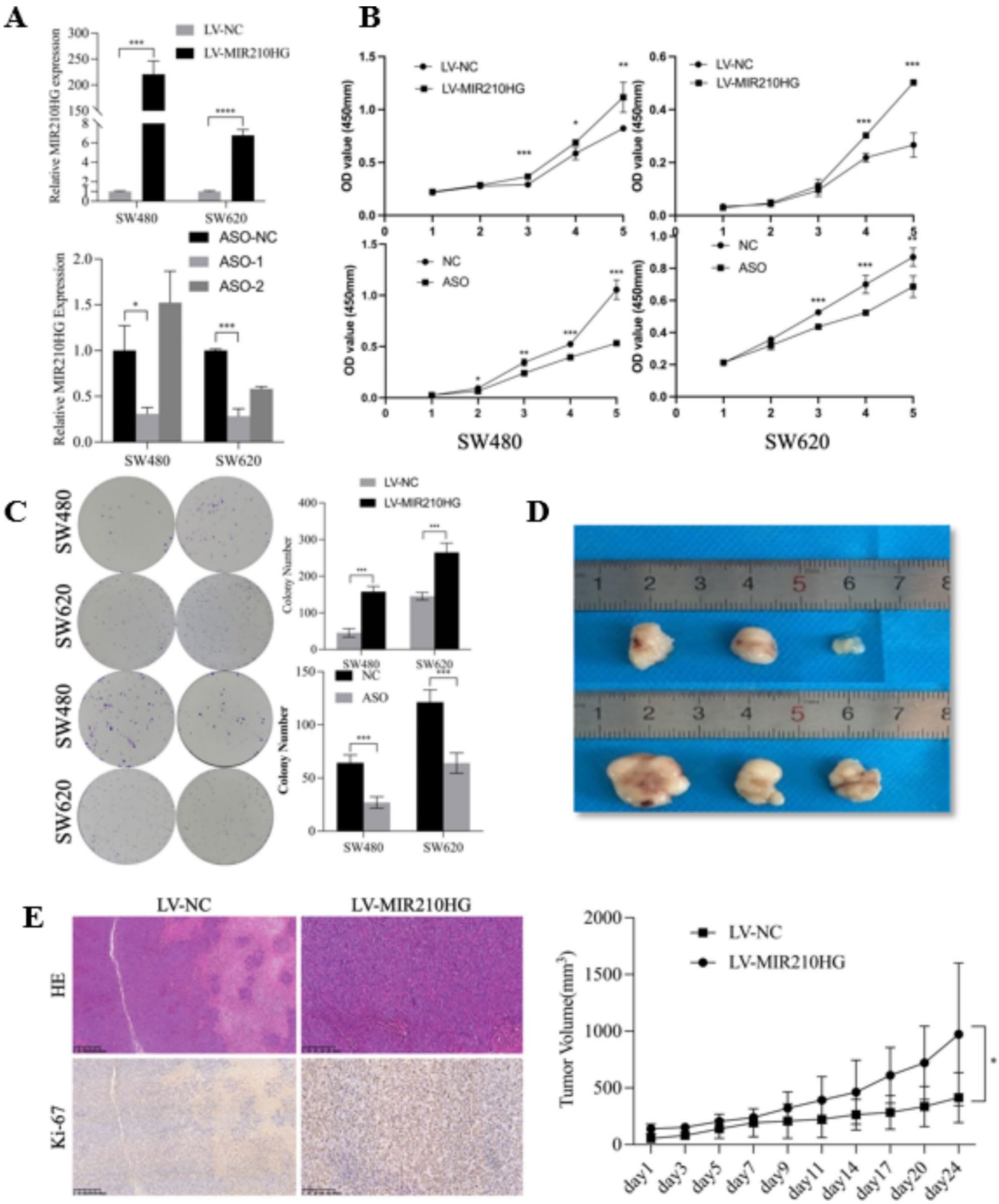

Fig. 2. The lncRNA MIR210HG promotes CC cell proliferation in vitro and in vivo. (**A**) SW480 and SW620 cells were transfected with the lentivirus LV-MIR210HG and ASO, and lncRNA expression was analyzed via qPCR. The data are presented as the average ± SD of three independent experiments. *$P < 0.05$, ***$P < 0.001$. (**B**) CCK-8 assays were used to detect the proliferation of SW480 and SW620 cells in which MIR210HG was overexpressed or knocked down. The data are presented as the average ± SD of three independent experiments. *$P < 0.05$, **$P < 0.01$, ***$P < 0.001$. (**C**) A plate colony formation assay was used to detect the colony formation ability of SW480 and SW620 cells in which MIR210HG was overexpressed or knocked down. The data are presented as the average ± SD of three independent experiments. ***$P < 0.001$. (**D**–**E**) Detection of changes in tumor morphology and volume, as well as HE and Ki-67 staining, in subcutaneous tumors from a xenograft mouse model of MIR210HG overexpression. *$P < 0.05$.

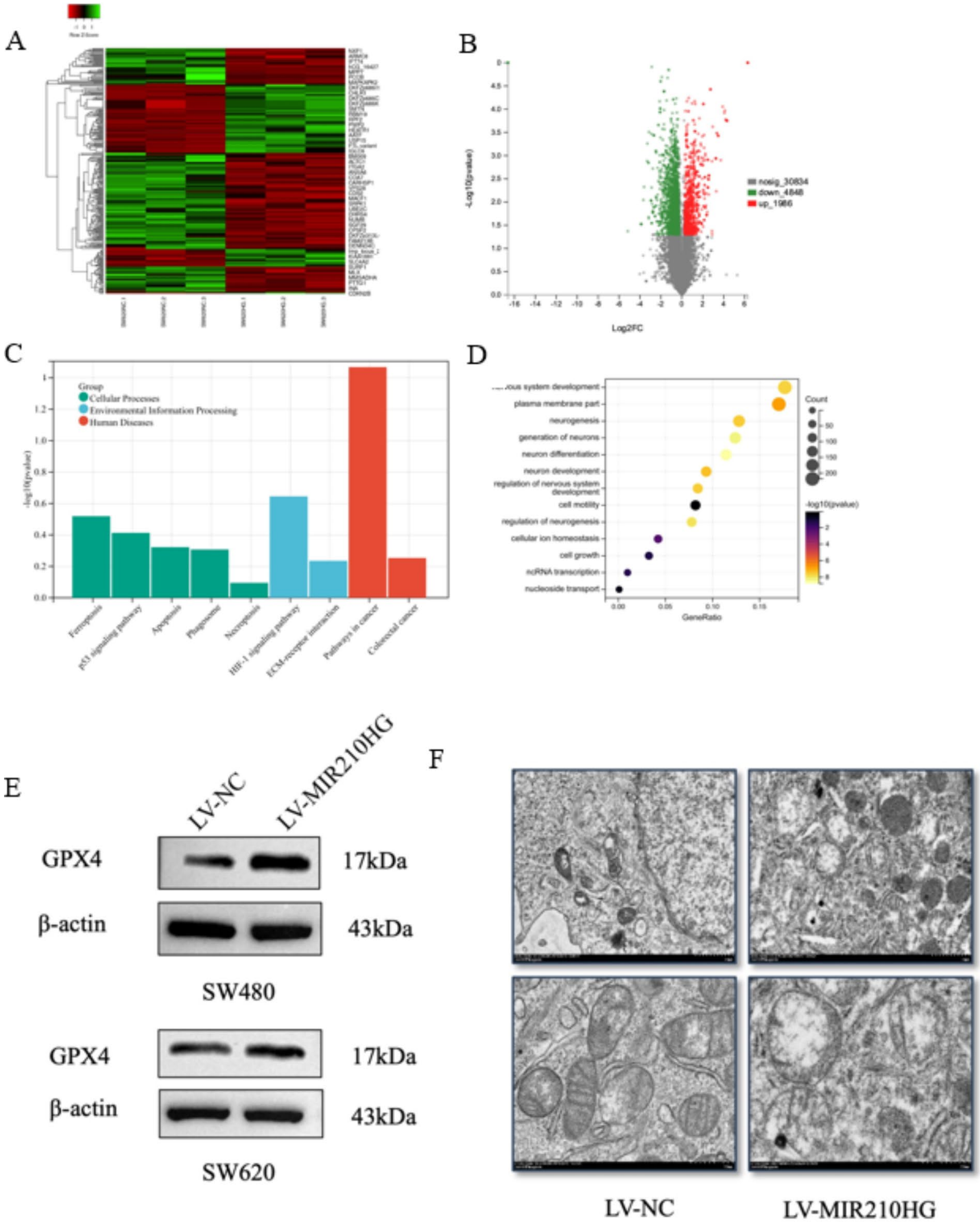

Fig. 3. Heatmap (**A**) and volcano plot (**B**) showing statistically significant changes in SW620 cells after MIR210HG overexpression. Gene Ontology (GO) (**C**) and Kyoto Encyclopedia of Genes and Genomes (KEGG) (**D**) pathway analyses of the differentially expressed genes associated with MIR210HG. Western blotting (**E**) was used to verify the effect of MIR210HG overexpression on ferroptosis-related RNA and protein expression. The grouping of blots cropped from different gels of same samples. TEM (**F**) showing the subcellular morphology of SW620 cells overexpressing MIR210HG and control cells.

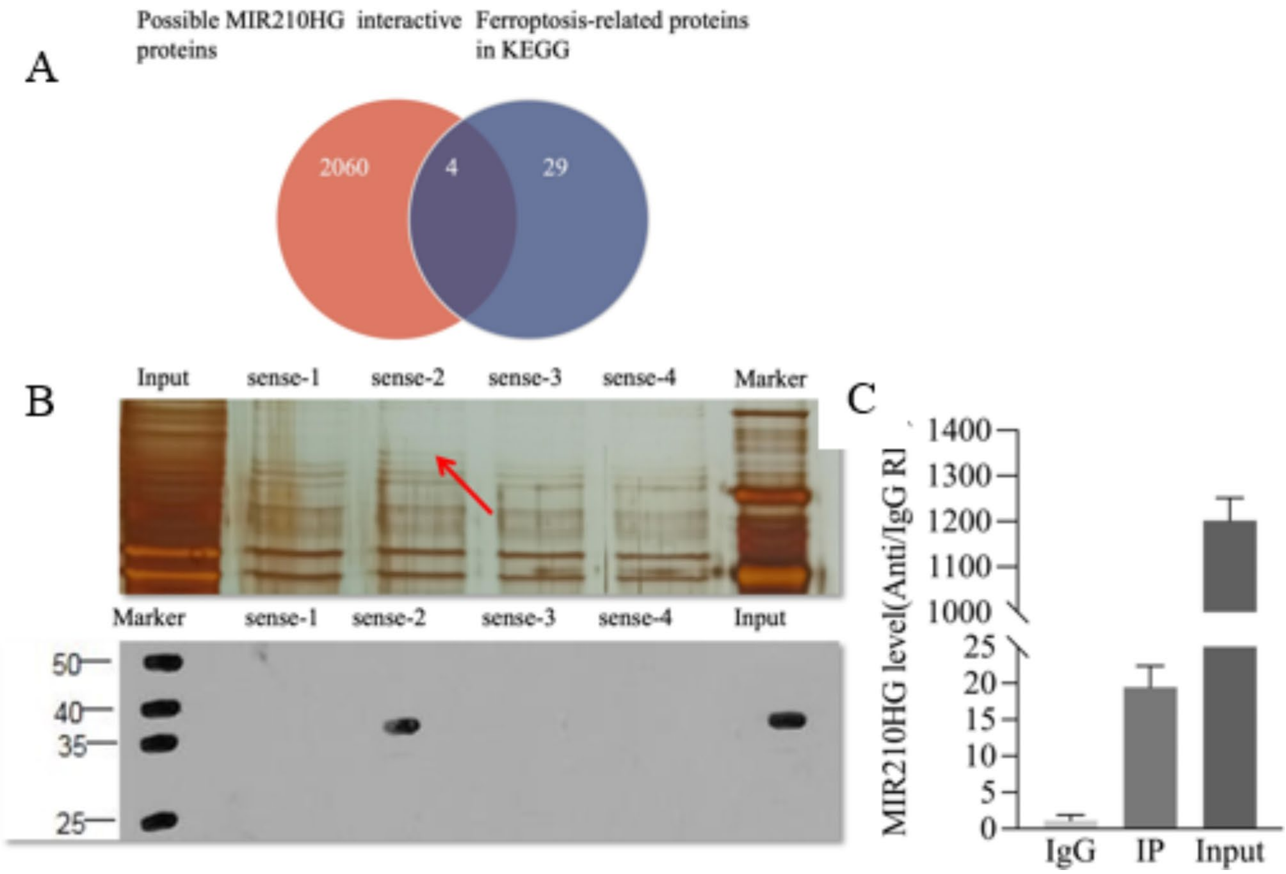

Fig. 4. The Venn diagram of downstream binding proteins (**A**, red) and ferroptosis pathway-related proteins (**A**, blue) predicted by catRAPID. In the RNA pull-down experiment (**B**), a differential band was observed in the sense-2 silver staining image, with the target protein PCBP1 detected in the sense-2 and input groups. RIP-qPCR ex-periment (**C**) showed an increase in the IP experimental group compared to the IgG control group. The data are presented as the average ± SD.

Discussion

This study provides convincing evidence that the upregulation of MIR210HG is significantly associated with poor survival in patients with CC. Our findings are consistent with previous research that implicated lncRNAs in the pathogenesis and progression of various malignancies, including CC. We observed that MIR210HG overexpression is linked to enhanced proliferation of CC cells both in vitro and in vivo, suggesting that MIR210HG may serve as an oncogenic driver in the context of CC.

The oncogenic potential of MIR210HG was further substantiated by its ability to inhibit the ferroptosis pathway, a form of regulated cell death characterized by the accumulation of iron-dependent lipid peroxidation products. Previous studies have highlighted the importance of ferroptosis in cancer biology, with evidence suggesting that its suppression can contribute to tumorigenesis[50].

Our discovery that MIR210HG overexpression leads to the upregulation of GPX4 and consequent suppression of ferroptosis aligns with findings by Yang et al.[51], who identified GPX4 as a crucial regulator of ferroptosis. This provides insight into the possible mechanisms by which MIR210HG may contribute to cellular proliferation and tumor growth. The substantial modulation of mitochondrial morphology in cells overexpressing MIR210HG reinforces the notion that MIR210HG plays a role in the suppression of ferroptosis, which was further validated by the upregulation of GPX4.

Our study also highlighted the binding affinity of MIR210HG for PCBP1, an iron chaperone protein that regulates iron homeostasis and mitigates oxidative stress within the cell[52]. The interaction between MIR210HG and PCBP1 suggests a novel regulatory axis that may explain the observed proliferative phenotype in CC cells. Previous studies have shown that in head and neck cancer, PCBP1 helps reduce cancer cell sensitivity to ferroptosis inducers through its dual function of repressing BECN1 and ALOX15 mRNAs[43]. By modulating the ferroptosis pathway, MIR210HG may enable CC cells to evade cell death and sustain malignant growth.

Our study highlighted the binding affinity of MIR210HG for PCBP1, an iron chaperone protein that regulates iron homeostasis and mitigates oxidative stress within the cell[8]. The interaction between MIR210HG and PCBP1 suggests a novel regulatory axis that may explain the observed proliferative phenotype in CC cells. By modulating the ferroptosis pathway, MIR210HG may enable CC cells to evade cell death and sustain malignant growth.

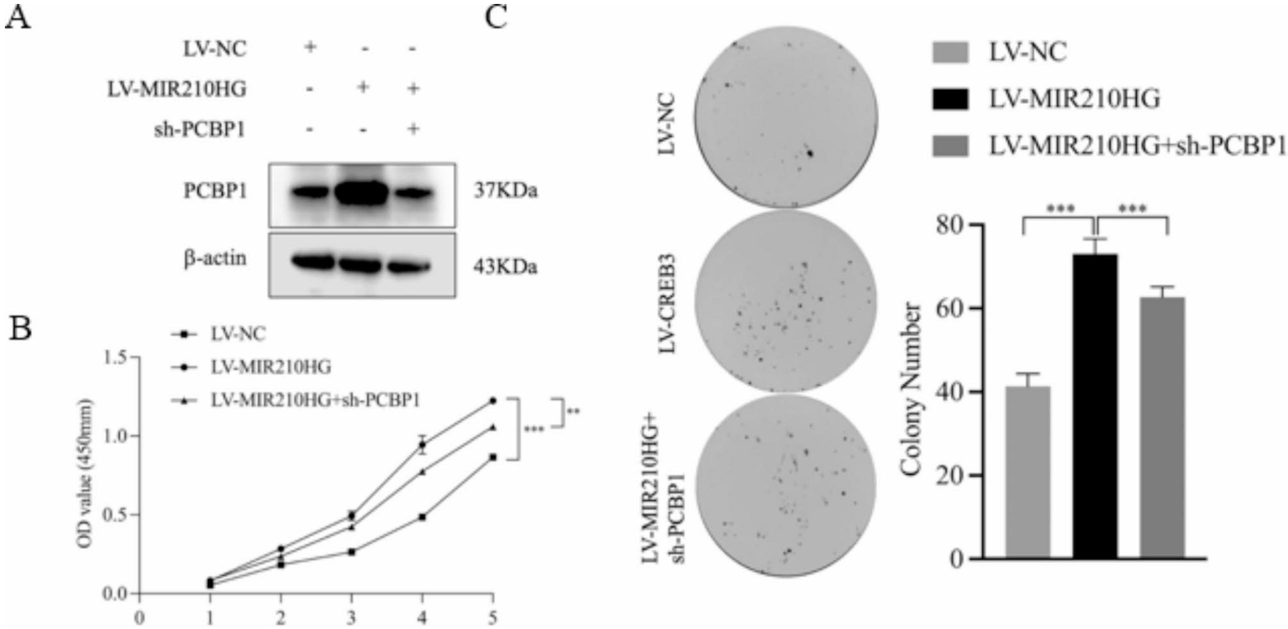

Fig. 5. Knockdown of PCBP1 counteracts the proliferative phenotype induced by overexpression of MIR210HG in colon cancer cells: SW480 cells were transfected with LV-NC, LV-MIR210HG, or LV-MIR210HG + sh-PCBP1. Among the three experimental groups, MIR210HG overexpression resulted in a significant upregulation of PCBP1 expression compared to that in the control group, while cotransfection with the two viruses led to a decrease in PCBP1 expression compared to that in the MIR210HG overexpression group (**A**). The results of the CCK-8 assay (**B**) and colony formation assay (**C**) demonstrated that knockdown of PCBP1 significantly attenuated the proliferative advantage conferred by overexpression of MIR210HG on colon cancer cells. The grouping of blots cropped from different gels of same samples. The data are presented as the average ± SD of three independent experiments. $^{**}P < 0.01$, $^{***}P < 0.001$.

The functional implications of MIR210HG in CC cell proliferation were further confirmed by our experiments showing that knockdown of PCBP1 attenuated the pro-proliferative advantage conferred by MIR210HG overexpression. This suggests that the pro-proliferative effects of MIR210HG are, at least in part, mediated through its interaction with PCBP1. Our data suggest that the MIR210HG-PCBP1 axis could represent a potential therapeutic target for the development of strategies aimed at inhibiting CC cell proliferation.

Our study is not without limitations. Although we demonstrated the overexpression of MIR210HG in CC and its association with survival outcomes, the nature of this study does not allow for the determination of a causal relationship. Additionally, while we have suggested that MIR210HG may serve as a potential biomarker for CC, further studies involving a larger cohort of patients are necessary to validate this potential.

In conclusion, our study advances the understanding of the role of MIR210HG in the pathobiology of CC and suggests that MIR210HG may serve as a novel target for therapeutic intervention. The MIR210HG-PCBP1 axis represents a compelling target for further investigation and has the potential to improve the prognosis and treatment of patients with CC.

Materials and methods

Cell culture and treatment

The human colon cancer cell lines HCT116, SW620, HT29, LOVO, and SW480 were obtained from the American Type Culture Collection (ATCC). The normal human colon epithelial cell line FHC was obtained from the National Collection of Authenticated Cell Cultures (NCAAC). All cancer cell lines were maintained in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal bovine serum (FBS), 100 U/ml penicillin, and 100 μg/ml streptomycin. FHC were cultured in DMEM supplemented with 10% FBS, 10 mM HEPES, 10 μg/ml insulin, 5.5 μg/ml transferrin, 0.5 μg/ml hydrocortisone, and 20 ng/ml human recombinant epidermal growth factor. Cultures were maintained at 37 °C in a humidified atmosphere of 5% CO_2.

Lentivirus and antisense oligonucleotide (ASO) transfection

Lentiviral vectors encoding MIR210HG and the corresponding negative controls (LV-NCs) were produced by Genechem Co., Ltd. Antisense oligonucleotides targeting MIR210HG (ASO-MIR210HG) and scrambled negative control ASOs were synthesized by RiboBio Co., Ltd. SW480 and SW620 cells were seeded in 6-well plates at a density of 1×10^5 cells/well and transduced with lentivirus or transfected with ASOs using Lipofectamine 3000 (Invitrogen) according to the manufacturer's instructions. The medium was replaced after 24 h, and cells were selected with 2 μg/ml puromycin to establish stable cell lines overexpressing MIR210HG.

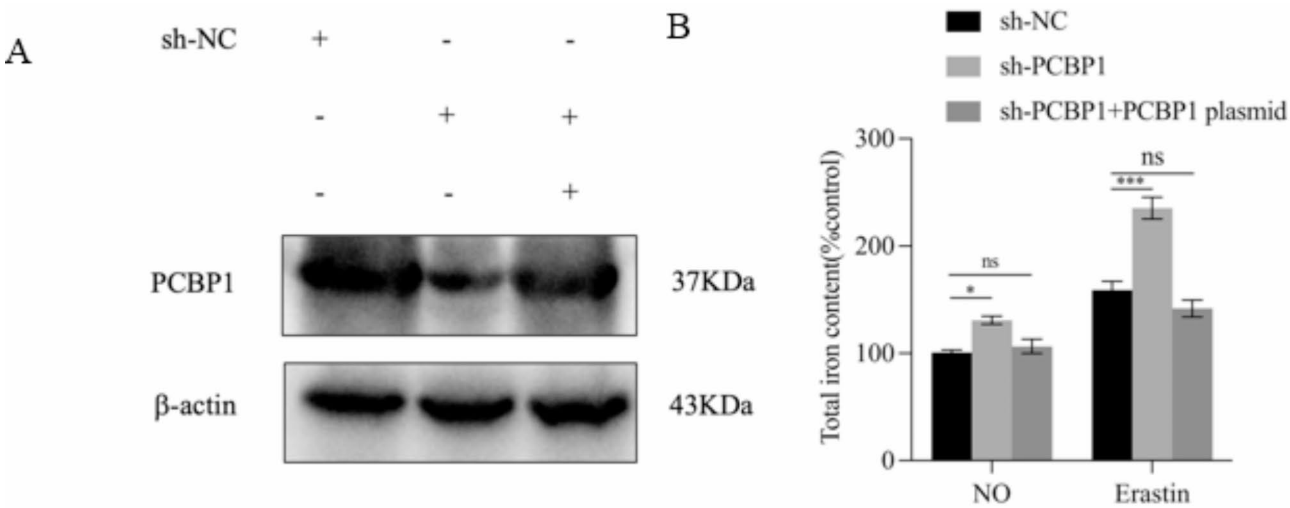

Fig. 6. PCBP1 modulates the intracellular iron content and affects the sensitivity of colon cancer cells to ferroptosis. (**A**) Western blot analysis showing the expression levels of PCBP1 in SW480 colon cancer cells in the control group, PCBP1-knockdown group, and PCBP1-restored expression group. Actin was used as a loading control. The PCBP1 expression in the knockdown group was significantly greater than that in the control group, while PCBP1 expression was restored in the re-expression group. The grouping of blots cropped from different gels of same samples. (**B**) Bar graph illustrating the measurement of the intracellular iron content. The x-axis represents the solvent control and erastin-treated groups, each containing the control, PCBP1-knockdown, and PCBP1-restored expression subgroups. Overall, the Eras-tin-treated groups displayed greater intracellular iron content than did the solvent control groups. Within these two main groups, the PCBP1-knockdown subgroup presented a significant increase in the intracellular iron content compared to that of the control subgroup, whereas there was no significant difference between the control and PCBP1-restored ex-pression subgroups. These results suggest that PCBP1 modulates the sensitivity of colon cancer cells to ferroptosis by regulating intracellular iron metabolism. The results are shown as the mean ± standard deviation (SD). *$p < 0.05$ and ***$p < 0.001$ indicate statistical significance between groups. ns, no statistically significant difference between the indicated groups.

Quantitative polymerase chain reaction (q-PCR)

Total RNA was extracted using TRIzol reagent (Invitrogen), and cDNA was synthesized using the PrimeScript RT reagent Kit (TaKaRa). q-PCR was performed using SYBR Green Master Mix (TaKaRa) on a LightCycler 480 Real-Time PCR System (Roche). The relative expression levels of MIR210HG were normalized to those of the housekeeping gene GAPDH using the $2^{-\Delta\Delta Ct}$ method.

Fragments Sequence.

Sense-1 F: taatacgactcactatagggcgcggggcccggggccggtcgggt R: acacacagcctttctcaggtgcaggt.

Sense-2 F: taatacgactcactatagggggtcttggggtccacacctgcagagcta.

R: tttttagagacgaggtttcaccatg.

Sense-3 F: taatacgactcactatagggtacaaaaattagctaggcatggtggtgg R: ctcctccatccacagccccga.

Sense-4 F: taatacgactcactatagggtaggcccctcatatcccaggcctgct.

R: tcacaatgtttaagaaaatttatagatttgtg.

Cell proliferation assays

The Cell Counting Kit-8 (CCK-8) assay was utilized to assess cell proliferation. Briefly, cells were seeded in 96-well plates at a density of 3 × 103 cells/well. At the designated time points, CCK-8 solution (Dojindo) was added to each well and incubated for 2 h at 37 °C. The absorbance at 450 nm was measured using a microplate reader. For colony formation assays, cells were seeded in 6-well plates at a density of 500 cells/well and allowed to grow for 14 days. Colonies were fixed with methanol, stained with 0.1% crystal violet, and counted. Only colonies with more than 50 cells were considered.

In vivo tumor xenograft model

Male BALB/c nude mice aged 4–6 weeks were purchased from the Animal Experiment Center of Air Force Medical University and housed under specific pathogen-free conditions. SW480 cells stably overexpressing MIR210HG were subcutaneously injected into the flanks of the mice (5×10^6 cells per mouse). Anesthesia was induced using isoflurane to minimize discomfort during the injection process. Tumor growth was monitored by measuring tumor length (L) and width (W) with calipers, and tumor volume was calculated using the formula $(L \times W^2)/2$. After 4 weeks, the mice were euthanized using CO_2 asphyxiation followed by cervical dislocation to ensure death. All methods are implemented in accordance with relevant guidelines and regulations. The research aligns with the Animal Research: Reporting In Vivo Experiments (ARRIVE) Guidelines.

Protein extraction and western blotting
Total protein was extracted using RIPA lysis buffer containing a protease and phosphatase inhibitor cocktail (Beyotime). Protein concentrations were determined using the BCA Protein Assay Kit (Pierce). Equal amounts of protein were separated by SDS-PAGE and transferred onto PVDF membranes. After blocking with 5% nonfat milk, the membranes were incubated with primary antibodies overnight at 4 °C, followed by incubation with HRP-conjugated secondary antibodies. Bands were visualized using enhanced chemiluminescence (ECL) reagents (Millipore) and quantified by densitometry.

Transmission electron microscopy (TEM)
SW620 cells stably overexpressing MIR210HG were fixed with 2.5% glutaraldehyde, postfixed with 1% osmium tetroxide, dehydrated in graded ethanol, and embedded in Epon. Ultrathin sections were cut, stained with uranyl acetate and lead citrate, and examined via TEM.

Statistical analysis
Statistical analyses were performed using GraphPad Prism 9.0 (GraphPad Software, Inc.). The data are presented as the mean ± standard deviation (SD). Comparisons between two groups were performed using Student's t test, while one-way ANOVA was used for multiple comparisons. A p value < 0.05 was considered to indicate statistical significance.

Data availability
The ferroptosis-related proteins used in this study were sourced from the KEGG database (https://www.kegg.jp). These proteins are detailed in Supplementary Table S2 (KEGG_Ferroptosis.xlsx), which is provided as part of the supplementary materials for this manuscript. Access to the KEGG database is subject to the terms and conditions set forth by Kanehisa Laboratories.

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Author contributions

Conceptualization, X.-Q.W. and L.H.; methodology, X.-Q.W. and L.H.; validation, X.-Q.W., A.-Q.F. and L.H.; formal analysis, A.-Q.F.; investigation, A.-Q.F.; resources, L.H.; data curation, A.-Q.F.; writing—original draft preparation, X.-Q.W., A.-Q.F. and L.H.; writing—review and editing, X.-Q.W. and L.H.; visualization, A.-Q.F.; supervision, L.H.; project administration, X.-Q.W. and L.H.; funding acquisition, L.H. All authors have read and agreed to the published version of the manuscript. A.-Q.F. contributed equally to this work.

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Declarations

Ethics approval and consent to participate

The gene expression detection of clinical tissue samples used in this study and the subcutaneous tumor formation experiment protocol in nude mice have been approved by the Ethics Committee of the First Affiliated Hospital of the Air Force Medical University.

Competing interests
The authors declare no competing interests.

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